ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

COMBINED SCIENCE

PAPER 2 Theory

4003/2

2 Hours

JUNE 2025 SESSION

Additional materials:
Answer booklet
Calculator (Optional)
String

The Periodic Table is provided on page 14.

Do not remove any sheets from the question paper.

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top.

Section A
Answer **all** questions.
Write your answers in the spaces provided on the question paper.

Section B
Answer any **two** questions.
Write your answers in the separate answer booklet provided.

Section C
Answer any **two** questions.
Write your answers in the separate answer booklet provided.

Section D
Answer any **two** questions.
Write your answers in the separate answer booklet provided.

For examiner's use	
Section A	
Section B	
Section C	
Section D	
TOTAL	

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question.

This question paper consists of 14 printed pages.

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Section A

Answer **all** questions in the spaces provided on the question paper.

1. **(a)** State any **two** functions of water in the body.

1 .....

2 .....

[2]

(b) **Fig.1.1** shows a graph of daily energy requirements for two groups of people.

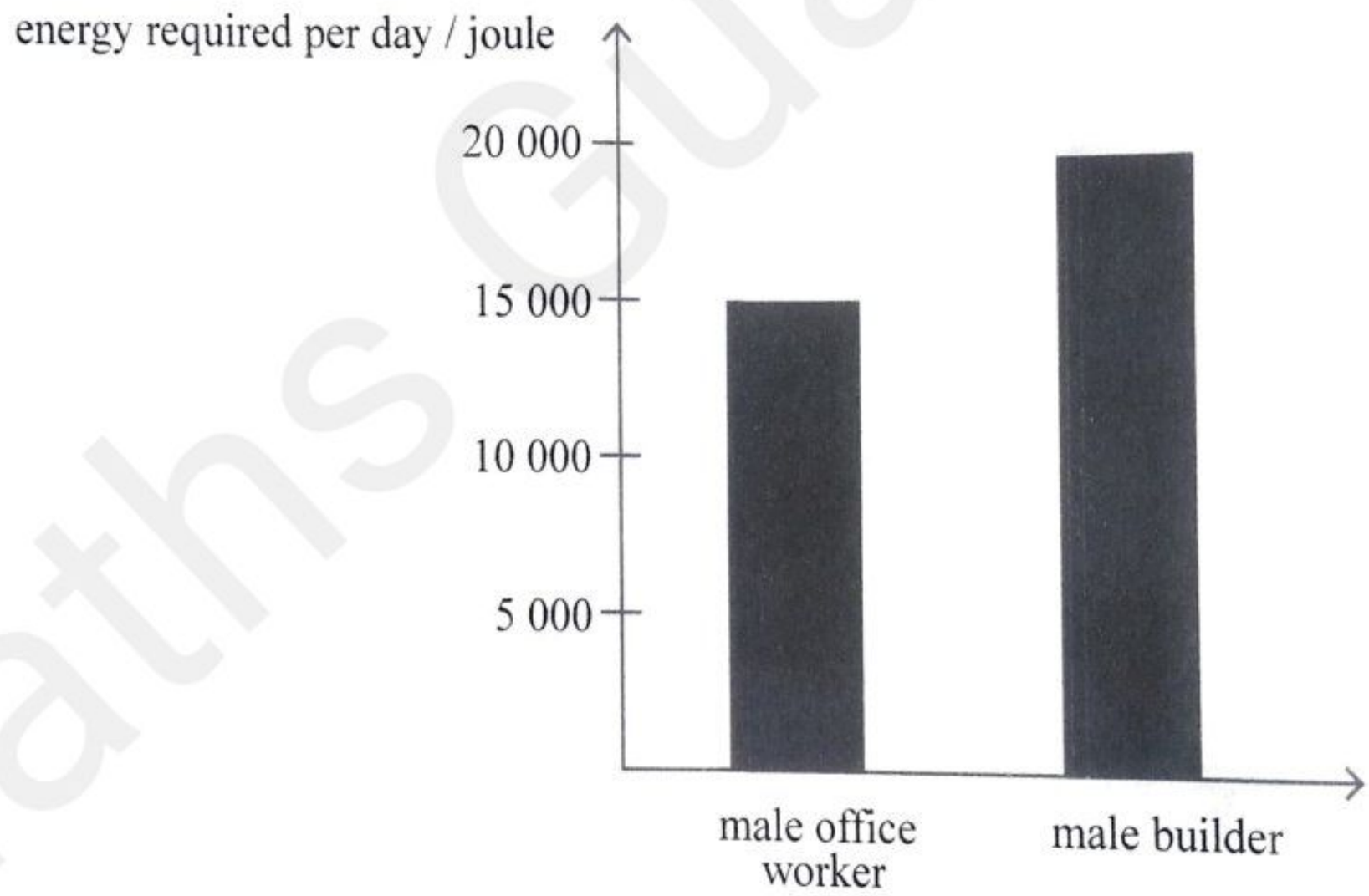

Fig.1.1

(i) With reference to **Fig.1.1**, explain the difference in energy requirements for a male office worker and a male builder.

.....

.....

[2]

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(ii) Identify any other consideration that can be made for a diet of different people.

.....

[1]

(c) State any **two** effects of malnutrition.

1 .....

2 .....

[2]

2. **(a)** Name the part of the male reproductive system that carries sperms to the urethra.

.....

[1]

(b) State a function of the epididymis.

.....

[1]

(c) Give the **two** functions of the testes.

.....

.....

[2]

(d) Name **one** part of the male reproductive system that produces seminal fluid.

.....

[1]

(e) Describe and explain the effect of gonorrhoea on the part named in **(a)**.

.....

.....

[2]

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3. (a) **Table 3.1** shows properties of the sub-atomic particles.

Table 3.1

sub-atomic particle	position	mass	charge
electron		$\frac{1}{1840}$	negative
neutron	nucleus		zero
proton	nucleus	1	

Complete **Table 3.1**.

[3]

(b) The oxygen element exists as either oxygen atom **A** or **B**. Atom **A** has a proton number of 8 and a mass number of 16 while atom **B** has a proton number of 8 and a mass number of 18.

(i) State the name used to describe the oxygen atoms **A** and **B**.

.....

[1]

(ii) Calculate the number of neutrons in atom **B**.

.....

[1]

(iii) Write the electronic configuration of atom **A**.

.....

[2]

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4. **Fig.4.1** shows the apparatus used in the electrolysis of water.

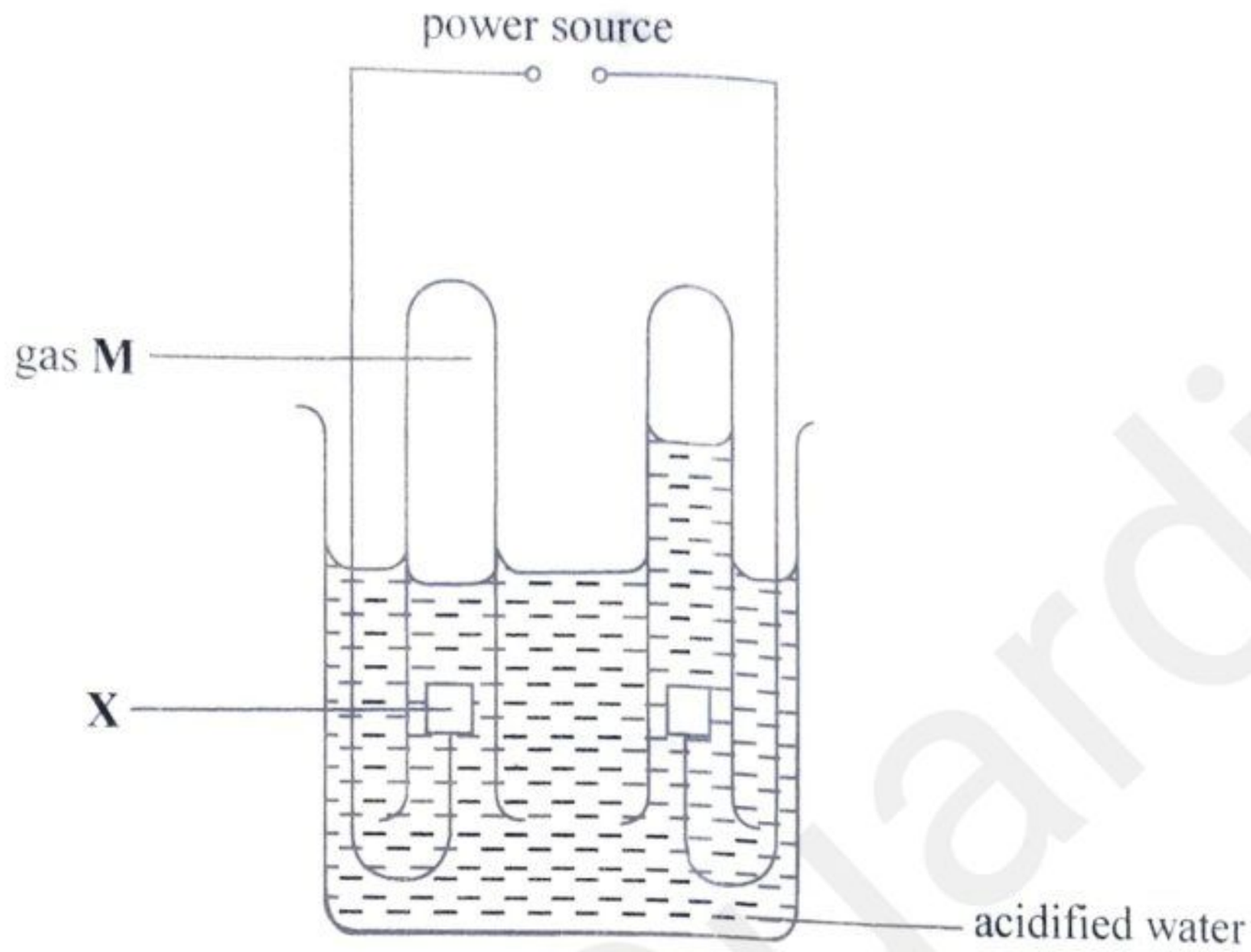

Fig.4.1

(a) Define the term *electrolysis*.

.....

.....

.....

[1]

(b) Name any **one** material used to make the electrodes..

.....

[1]

(c) Identify gas **M**.

.....

[1]

(d) Explain why the water is acidified?

.....

[1]

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(e) Describe, with the aid of an equation, the reaction that takes place at **X**.

[3]

5. (a) **Fig.5.1** shows a carburetor.

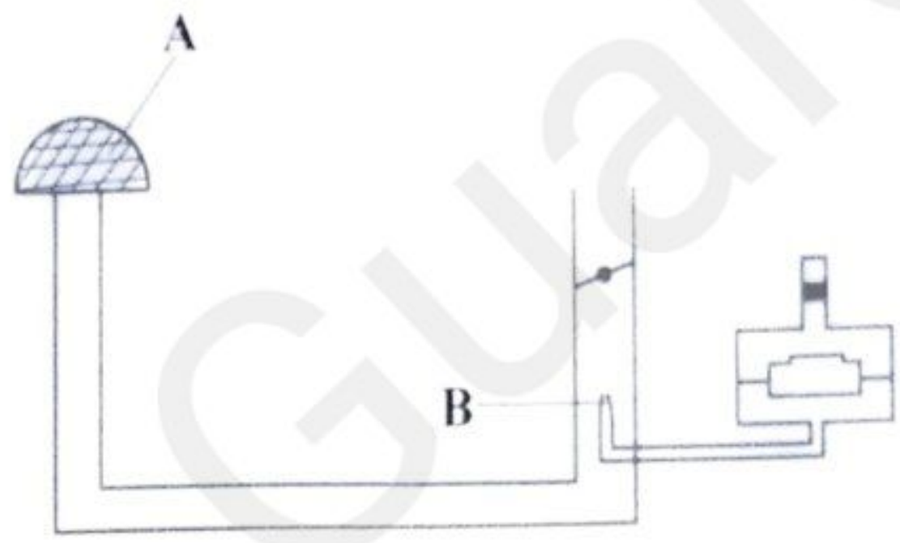

Fig.5.1

(i) Identify the parts labelled **A** and **B**.

A

B

[2]

(ii) State the effect of a blockage of part **B**.

[1]

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(b) State any **two** differences between the intake and exhaust strokes of a petrol engine.

.....

.....

.....

.....

[2]

(c) State any **two** advantages of modern fuel engines over old models.

.....

.....

.....

[2]

6. **(a)** State **two** measuring instruments used in an experiment to verify Ohm's law.

instrument 1 .....

instrument 2 .....

[2]

(b) A resistor of 5 Ω is connected in a circuit. A current of 0.5 A flows in the circuit.

Calculate the voltage across the resistor.

[3]

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Section B

Answer any **two** questions on the separate answer booklet provided.

7. **(a)** Write a word equation to describe the process of food manufacturing in green plants [2]

(b) State and explain how any **two** factors affect the rate of photosynthesis. [4]

(c) Describe any **two** ways in which photosynthesis is important. [4]

8. **(a)** **(i)** Explain any **two** problems caused by low species diversity. [4]

(ii) Describe the advantage of a food web over a food chain. [1]

(b) **Fig.8.1** shows part of the nitrogen cycle.

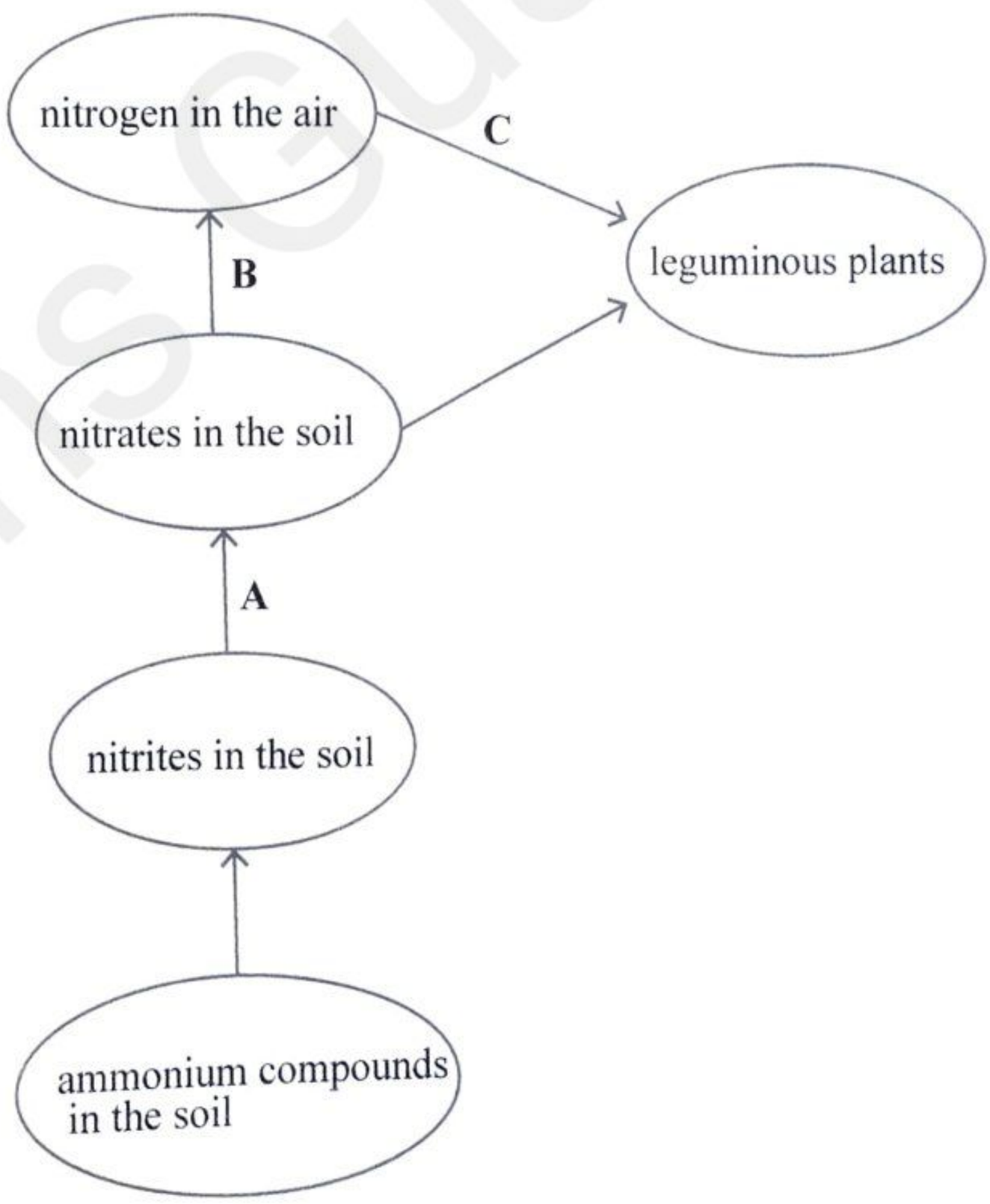

Fig.8.1

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(i) Identify the **three** groups of bacteria that are responsible for processes **A**, **B** and **C**. [3]

(ii) State the effect of process **B** on the soil. [1]

(iii) Describe how animals obtain nitrogen compounds. [1]

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9. **(a)** **(i)** State the organism that causes bilharzia. [1]

(ii) Give any **two** methods of transmission of bilharzia. [2]

(iii) Describe the life cycle of the bilharzia parasite. [4]

(b) State **two** differences between passive and active immunity. [2]

(c) Newly born babies are given the BCG vaccine.

Name a disease that is prevented by the BCG vaccine. [1]

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Section C

Answer any **two** questions on the separate answer booklet provided.

10. **(a)** Write a word equation for the reaction between calcium carbonate and hydrochloric acid. [2]

(b) State, with a reason, the precaution to be taken when handling the reactants in **(a)**. [2]

(c) Give the pH of the products. [1]

(d) Suggest a way of separating the products. [1]

(e) Describe a test for the gas produced. [2]

(f) State any **two** industrial uses of the gas. [2]

11. **(a)** **(i)** State any **two** raw materials which are needed for the production of biogas. [2]

(ii) Give any **two** optimum conditions necessary for the production of biogas. [2]

(b) Describe how biogas is produced. [3]

(c) Give any **two** gases that are components of biogas. [2]

(d) State any **one** use of biogas. [1]

12. **(a)** **(i)** State any **two** properties of chlorine. [2]

(ii) Describe any **two** differences between chlorine and bromine. [2]

(iii) State any **two** uses of chlorine. [2]

(b) Calculate the number of moles in 3.01×10^{23} molecules of chlorine given that the Avogadro constant is 6.02×10^{23}. [2]

(c) Describe any **two** chemical properties of Group I elements. [2]

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Section D

Answer any **two** questions on the separate answer booklet provided.

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13 **(a)** **Fig.13.1** shows an inclined plane used to lift a load of mass 16 kg. [$g = 10\ m/s^2$]

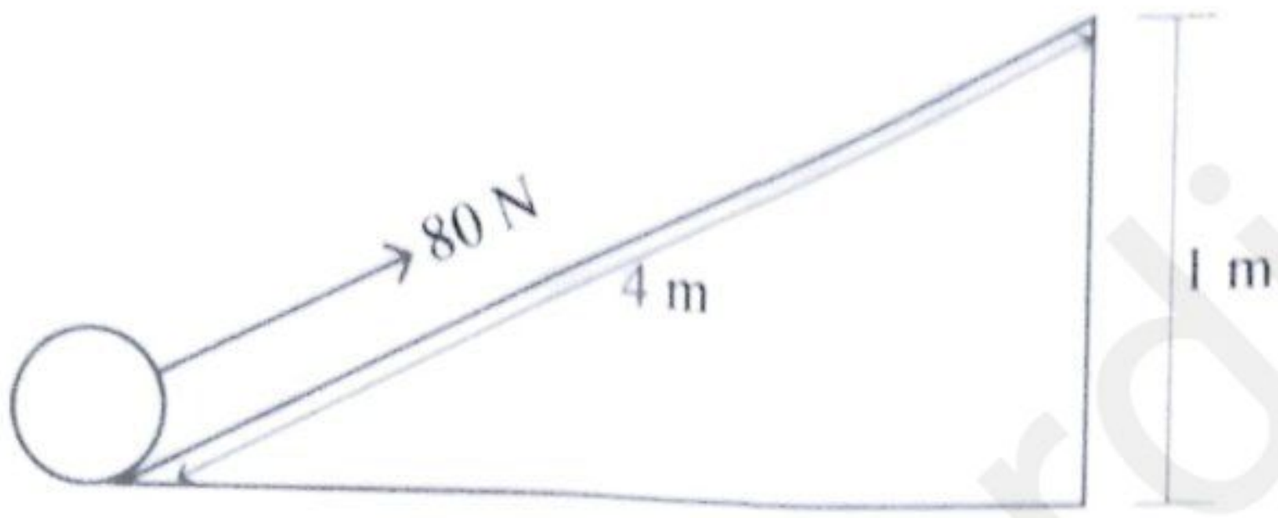

Fig.13.1

(i) Calculate the velocity ratio (V.R) of the inclined plane. [2]

(ii) Calculate the mechanical advantage (M.A) of the inclined plane. [3]

(iii) State any **two** ways of improving the efficiency of an inclined plane. [2]

(b) **Fig.13.2** shows gears that are used in a machine.

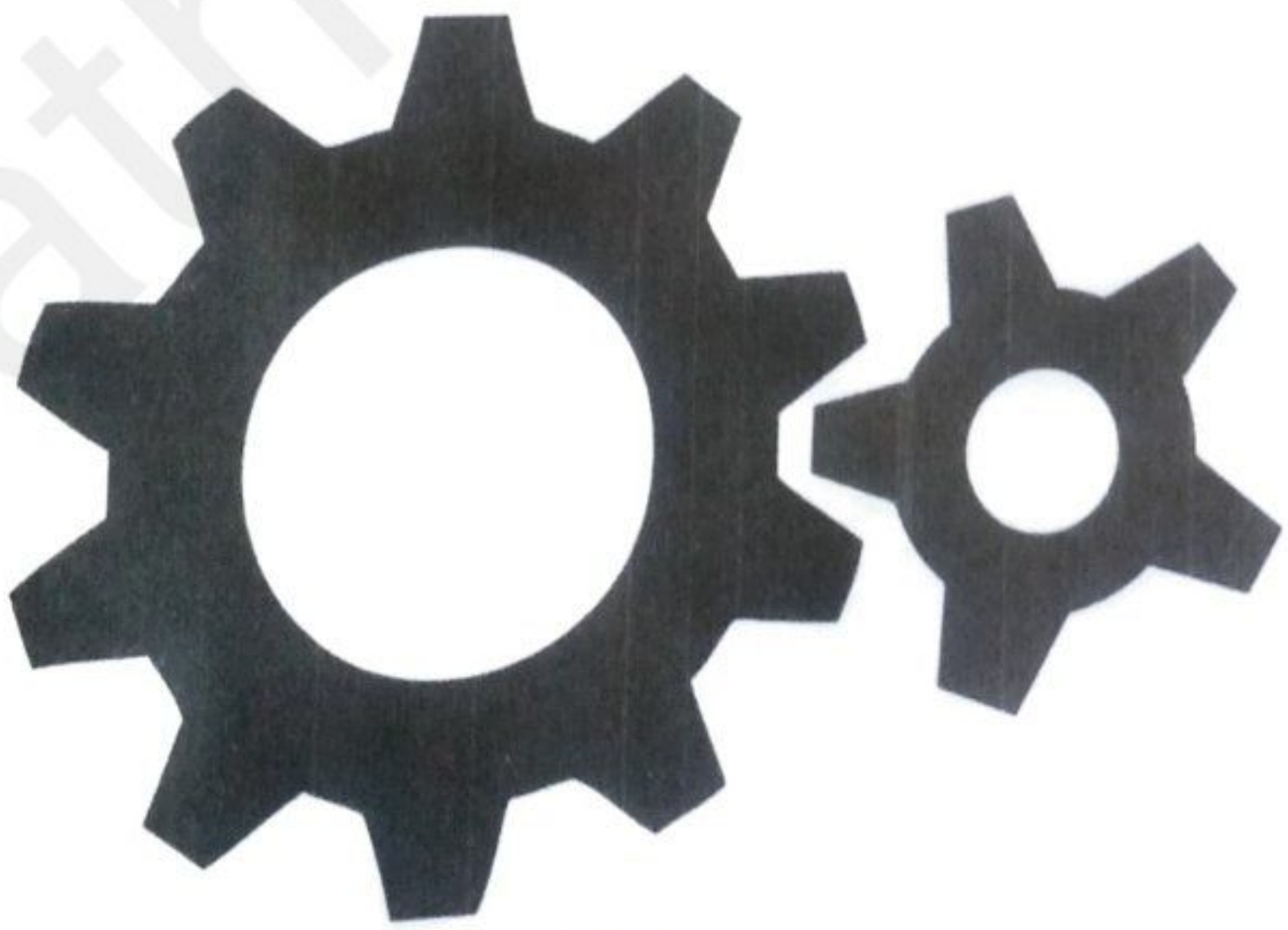

Fig.13.2

(i) Calculate the velocity ratio of the machine. [2]

(ii) State any **one** machine that uses gears. [1]

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14. **(a)** **(i)** Give any **two** sources of renewable energy. [2]

(ii) Give any **one** form of energy which is produced by lightning. [1]

(b) Energy is converted when a torch is switched on.

(i) Describe the energy changes that occur when the torch is switched on. [2]

(ii) Identify the energy converter in the torch. [1]

(c) **(i)** Identify the form of energy possessed by a stretched catapult. [1]

(ii) State any **one** form of energy that is produced when the catapult is released. [1]

(d) **Fig.14.1** shows solar panels installed on a house roof.

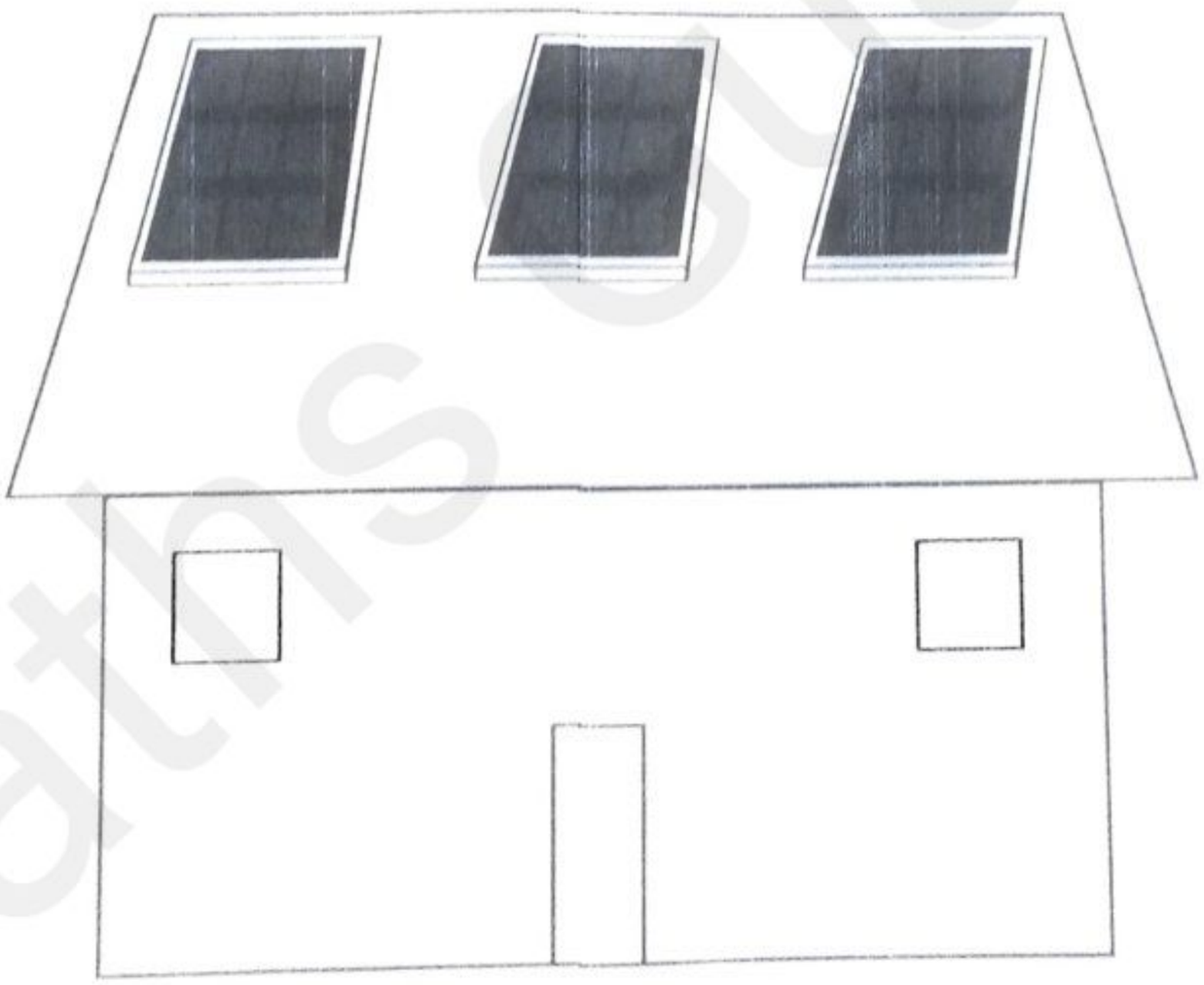

Fig.14.1

(i) State the source of energy captured by the solar panels. [1]

(ii) State the type of connection that results in maximum voltage being produced by the three solar panels. [1]

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15. **(a)** Forces can be grouped into contact and non-contact forces.

Give any **two** examples of non-contact force. [2]

(b) **Fig.15.1** shows a block being pushed on the floor.

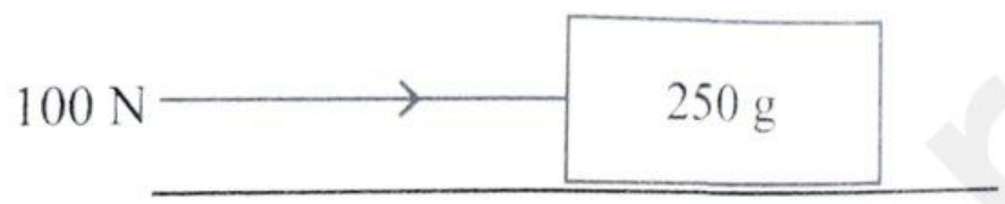

Fig.15.1

(i) Calculate the acceleration of the block. [3]

(ii) State a force which may reduce the acceleration of the block. [1]

(iii) State any **two** useful applications of the force stated in **b(ii)**. [2]

(c) State Newton's first law of motion. [2]

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